ADITYA CHUNDURI

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chunduri-aditya.github.io/Portfolio

ML/AI Engineer | Production systems across audio, vision, and language | End-to-end model design to deployment

Education

University of Southern California — Los Angeles, CA
Master of Science in Applied Data Science — *January 2024 – December 2025*
Relevant Coursework: Machine Learning, Deep Learning, NLP, Statistical Computing, Data Engineering

SRM Institute of Science and Technology — Chennai, India
Bachelor of Technology in Computer Science and Engineering (AI/ML Specialization) — *June 2019 – July 2023*

Technical Skills

Languages: Python, C++, SQL, JavaScript, Bash
ML/AI Frameworks: PyTorch, TensorFlow, Scikit-learn, Hugging Face, Keras, OpenCV, Librosa
MLOps and Cloud: Docker, AWS, FastAPI, Flask, Git, GitHub Actions, CI/CD, Gradio, LangChain
Data and Tools: PostgreSQL, ChromaDB, Neo4j, Pandas, NumPy, Plotly, Demucs

Experience

USC – Viterbi School of Engineering — Los Angeles, CA
Research Assistant, Computer Vision and Medical Imaging — *August 2024 – December 2024*
- Spearheaded CVD-Masks pipeline for retinal artery-vein classification using U-Net architectures, achieving 94% Dice score; presented progress in weekly team reviews.
- Streamlined medical image scraper with quality-validation controls, cutting manual curation time by 40% across 10,000+ images; shared reusable modules with collaborators.
- Designed modular PyTorch training pipelines with experiment tracking, reducing iteration cycles by 30% across 3 concurrent research projects.

SSN College of Engineering — Remote
Research Intern, Computer Vision and Object Detection — *June 2021 – July 2021*
- Delivered real-time object tracking pipelines at 30+ fps for automated video dataset generation, eliminating 100% of manual frame-annotation overhead.
- Fine-tuned YOLO-based CNN architectures, improving detection robustness by 15% across 5 environmental conditions; documented findings for team knowledge transfer.
- Produced 10+ reusable deep learning modules in TensorFlow and PyTorch, standardizing deployment workflows and reducing onboarding time for new contributors.

Projects

AI RemixMate: AI-Powered DJ Mixing Engine — September – December 2025
Python, PyTorch, Librosa, Demucs, Gradio
- Engineered end-to-end audio ML pipeline: stem separation (Demucs), spectral analysis, and Camelot Wheel harmonic matching for tempo-synchronized, key-compatible mix generation.
- Shipped Gradio web UI supporting 8+ audio formats with real-time BPM, key, and energy analysis; open-sourced with 9 GitHub stars within 3 months of release.

AkashicTree: Automated Content Generation Pipeline — June – August 2025
Python, Diffusers, Ollama, ElevenLabs, FLUX.1
- Conceived and shipped a model-agnostic Automated Content Generation (ACG) pipeline producing social posts across text, image, and audio from a single brief, cutting production time by 80%.
- Configured tiered backend-switching between local Ollama models and cloud APIs (GPT-4o-mini), reducing per-iteration inference costs by over 90% during development cycles.

AI Health Journal: Privacy-First LLM Assistant — January – May 2025
Python, Flask, Ollama, ChromaDB
- Independently deployed a local-first Retrieval-Augmented Generation (RAG) system with 7B-parameter LLM inference via Ollama and zero external API dependencies.
- Optimized ChromaDB vector retrieval for sub-second context injection across 10,000+ document chunks; applied Direct Preference Optimization (DPO) for health response alignment.

Model-Behavior-Lab: LLM Benchmarking Platform — September – December 2024
Python, Ollama, Plotly, CI/CD
- Launched end-to-end LLM evaluation platform running 5,000+ automated reasoning and hallucination tests across 10+ models with zero manual review.
- Structured JSON test suite with CI/CD integration so new models onboard and benchmark automatically, cutting evaluation setup from hours to under 5 minutes.

Publication

Wind Power Analysis Using Machine Learning – IJRASET, August 2023 (Co-Author): Co-developed a neural network framework integrating numerical weather data for wind energy forecasting across multiple datasets.